

CHAPTER 2 PRESENTATION

Mastering GIMP

An interactive and structured guide to GNU Image Manipulation Program, essential color tools, layer concepts, and advanced selection techniques.

What is GIMP?

An Open Source Creative Suite

GNU Image Manipulation Program (GIMP) is a powerful, multi-platform image editor ideal for image composition, creation, and professional photo retouching.

- **Free Access:** Zero-cost software with rich professional capabilities, supported across Windows, OS X, and Linux systems.
- **Expandable Power:** Supports online plugins and custom scripts to add fresh effects and advanced dynamic filters effortlessly.

GIMP Strengths

It offers paint suites, advanced scripting, customizable layers, and robust memory management that operates based on available disk size.

 GIMP Logo Wilbur Mascot on transparent background

Types of Digital Images

Bitmap (Raster) Images

Composed of a rectangular grid of pixels, where each pixel has a defined color value.

 **Limitation:** They lose sharpness and become pixelated when stretched or scaled up.

 **Best For:** Complex photographs and rich digital graphics.

Vector Images

Defined by geometric curves, points, and mathematical paths rather than pixels.

 **Advantage:** Can scale infinitely without losing quality or becoming blurred.

 **Best For:** Logos, fonts, icons, and precise technical artwork.

Supported File Formats

Format Extension	Full Name	Primary Characteristics & Usage
<code>.xcf</code> NATIVE	eXperimental Computing Facility	GIMP's native format. Preserves layers, transparency, channels, and edit histories.
<code>.jpg / .jpeg</code>	Joint Photographic Experts Group	Excellent for complex photographs. Uses compression which discards raw data to reduce size.
<code>.png</code>	Portable Network Graphic	Superb for web images. Fully supports transparency (alpha channels) without compression loss.
<code>.gif</code>	Graphics Interchange Format	Limited to 256 colors. Best for small icons, simple graphics, and web animations.
<code>.eps</code>	Encapsulated PostScript	A versatile format frequently used for high-fidelity vector-based images.

Starting GIMP & Workspace



The Toolbox

The control center containing drawing tools, color swatches, selectors, and structural transformation features to alter image shapes.



Tool Options

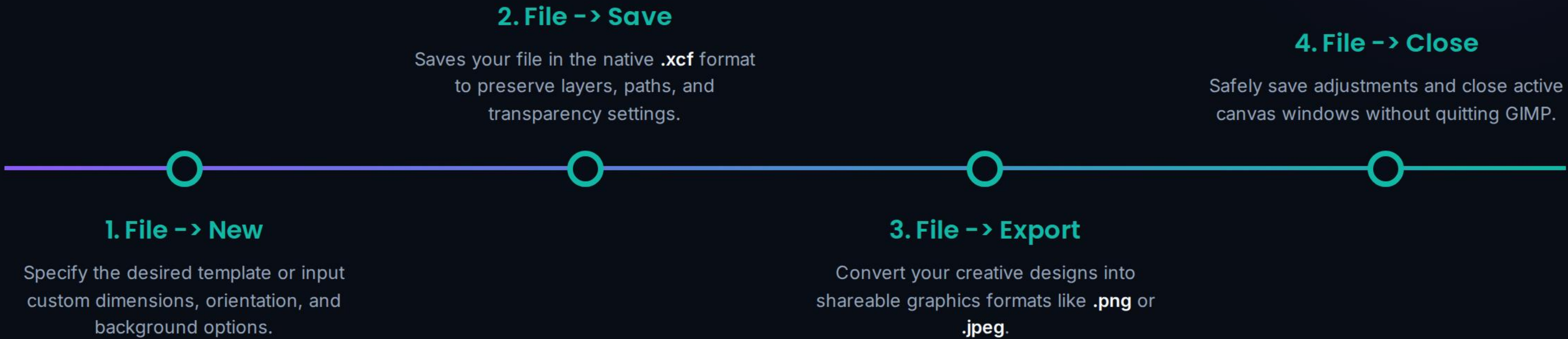
A dynamic panel located directly beneath the Toolbox. It displays customizable properties for whichever tool is currently active.



Image Window

The main workspace and canvas. It houses the image files you open and displays your active design project clearly.

The GIMP File Lifecycle



1. File -> New

Specify the desired template or input custom dimensions, orientation, and background options.

2. File -> Save

Saves your file in the native **.xcf** format to preserve layers, paths, and transparency settings.

3. File -> Export

Convert your creative designs into shareable graphics formats like **.png** or **.jpeg**.

4. File -> Close

Safely save adjustments and close active canvas windows without quitting GIMP.

Applying Colors & Gradients

Bucket Fill Tool (Shift + B)

This tool fills an active selection with a solid color of your choosing. It has three distinct modes:

FG Color Fill: Fills selection with current foreground color.

BG Color Fill: Fills selection with current background color.

Pattern Fill: Uses a selection of textures to fill spaces.

Blend Tool / Gradient (L)

Fills the chosen area with a smooth gradient transition between the selected foreground and background colors.

→ **Click-and-Drag:** Click on the canvas and drag in any direction to draw the gradient vector line.

● Adjust length and direction to change how colors blend.

Basic Selection Tools



Rectangle Select (R)

Designed to capture rectangular or square regions on the active canvas. Press and drag diagonally to draw the frame.



Ellipse Select (E)

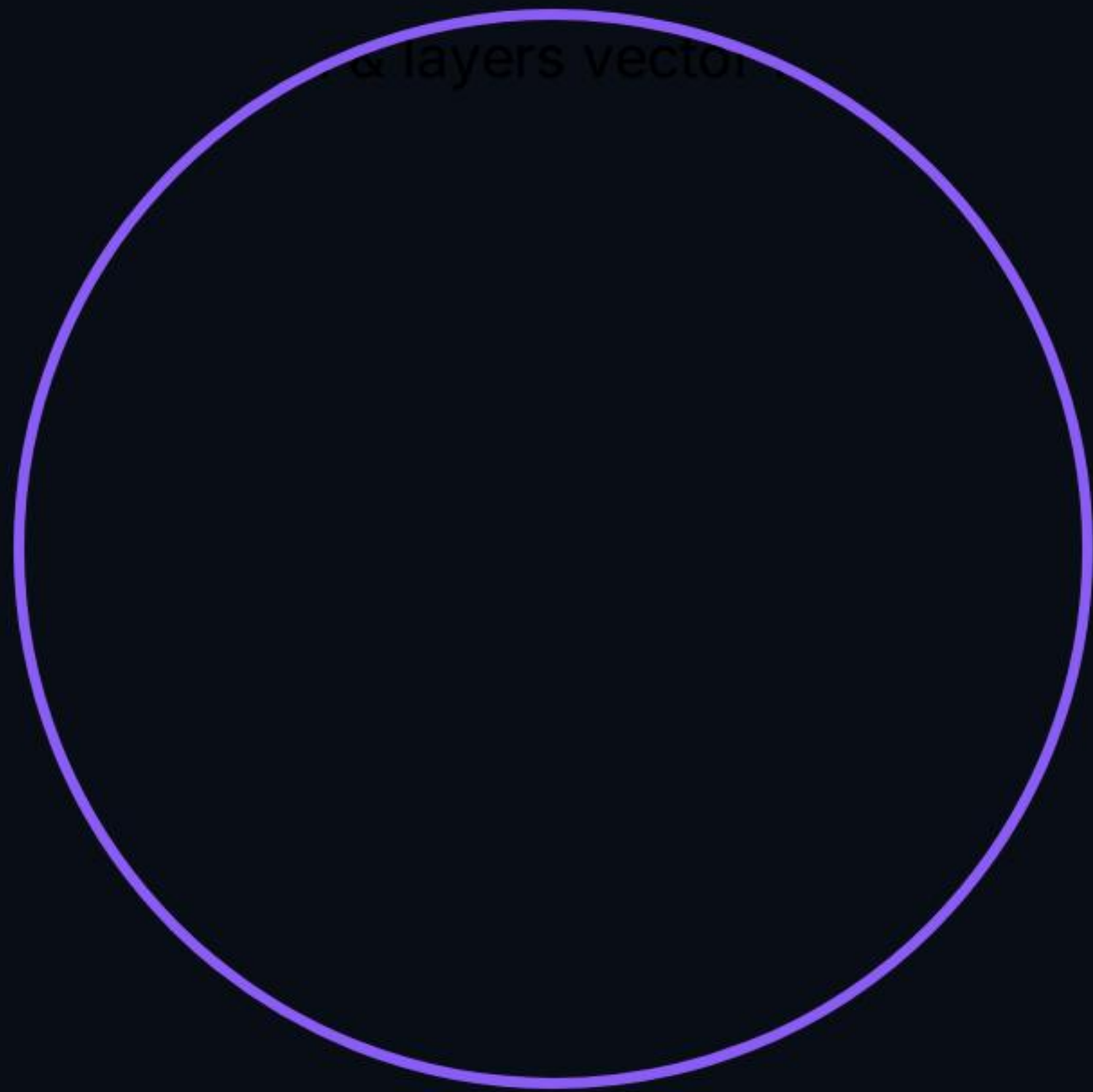
Captures perfectly circular or elliptical areas. Crucial when focusing on rounded elements of a graphic or photo.



Free Select / Lasso (F)

Allows freehand selector drawing. Move the pointer across the canvas to define custom boundaries of any shape.

Smart Selection Tools



Fuzzy Select vs Select By Color

When basic shapes fail, GIMP provides advanced, color-sensitive tools to quickly isolate targets:

Fuzzy Select (Magic Wand Tool): Selects contiguous or connected areas of a single color. It looks at neighboring pixels and expands until the color changes dramatically.

Select By Color Tool: Selects *all* regions on the canvas that share color similarities, regardless of whether those areas are connected.

The Scissors Select Tool

100%

Intelligent Edge Detection

Isolating Complex Objects

The **Scissors Select Tool (I)** is perfect for selecting shapes with high-contrast, distinct edges (such as isolating a building or a landmark from its background).

🟢 **Click Anchors:** Add dots along the object boundary. The tool automatically detects contrast differences and snaps the line to the edge.

🔗 **Interactive Curve:** Drag the paths between anchors to align them precisely before pressing Enter to convert to a selection.

The Magic of Layers

Layers are transparent sheets stacked on top of each other. They allow you to build complex designs without merging your original objects.

Isolation: Modify, paint, or scale one layer without affecting elements underneath.

Visibility: Toggle the "Eyeball" icon on and off to hide or display specific elements instantly.

Order: Raise or lower layers to change who sits on top of whom in your final image.

💡 Pro-Tip for Compositing

Keep a clean layer structure by renaming each layer according to its content (e.g., 'Background', 'Main Object', 'Text').

Editing & Refining Selections

-  **Scale Tool (Shift + S):** Used to enlarge or reduce the size of selected regions or entire layer graphics. Drag the handles or type numerical proportions in the popup dialog.
-  **Zooming Methods:** Enhance precision editing. Use **Ctrl + Mouse Scroll** (Scroll Up to zoom in, Scroll Down to zoom out) or load the **Navigation Dialog**.
-  **Feathering Selections:** Softens the hard boundary lines of a cutout. It creates a subtle, transparent gradient bleed at the selection edge, helping objects merge into new backgrounds seamlessly.

Questions?

You are now ready to jump into the lab activities and practice editing in GIMP!

Chapter 2: GNU Image Manipulation Program | Classroom Presentation

Image Sources



<https://upload.wikimedia.org/wikipedia/commons/thumb/0/02/Gimp-wilber-scalable.svg/1280px-Gimp-wilber-scalable.svg.png>

Source: commons.wikimedia.org



Thumbnail https://sm.pcmag.com/pcmag_me/comparison/a/adobe-illu/adobe-illustrator-vs-photoshop-whats-the-difference_vu5k.jpg

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